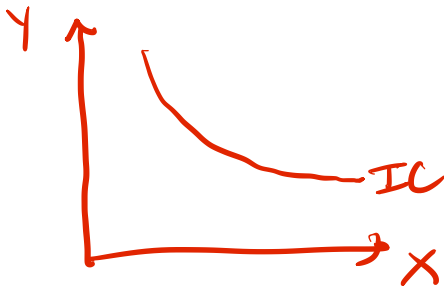


same utility

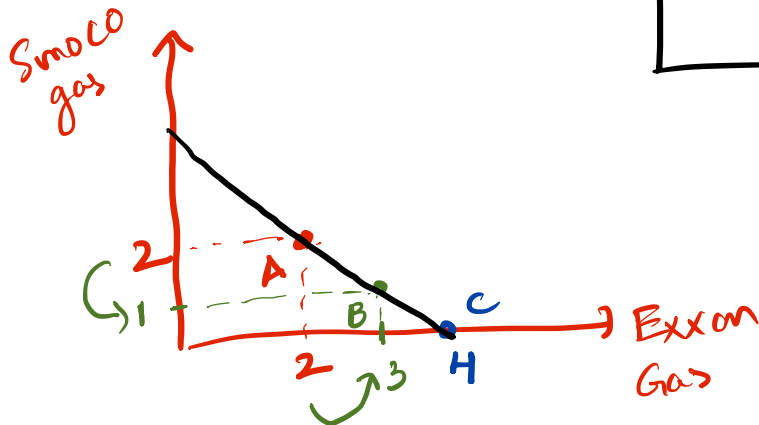
Indifference Curves



Now we will explore how the shape of the IC varies
by different types of goods...

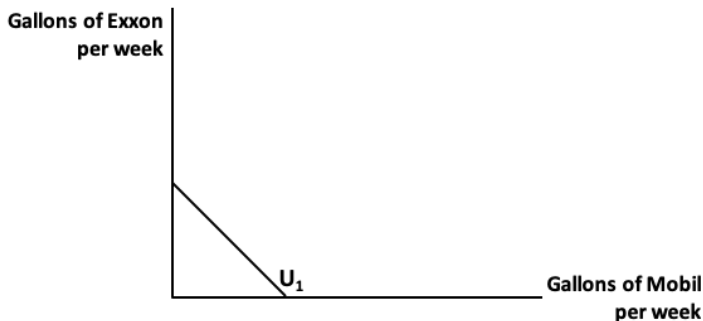
Perfect Substitutes

Two goods that plausibly serve identical functions



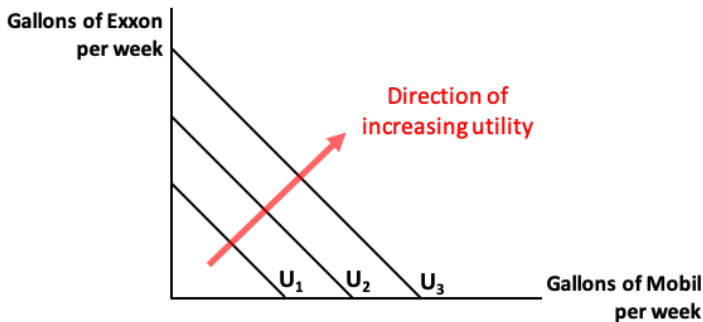
Perfect Substitutes

Two goods that plausibly serve identical functions.



Perfect Substitutes

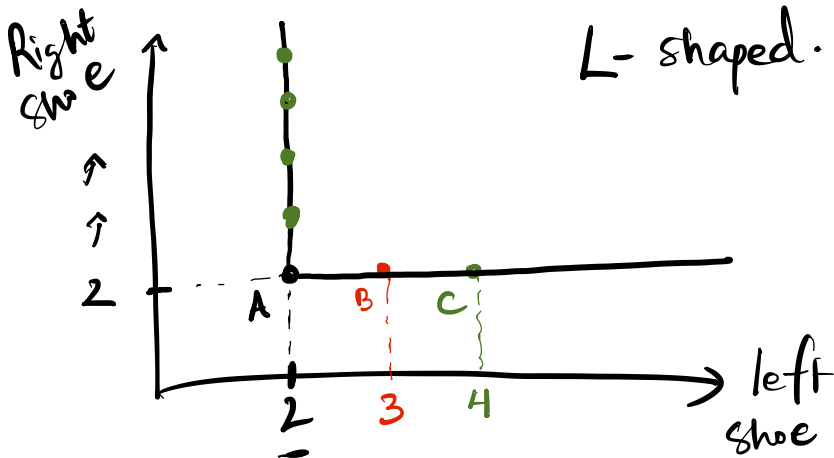
Two goods that plausibly serve identical functions.



Perfect Complements

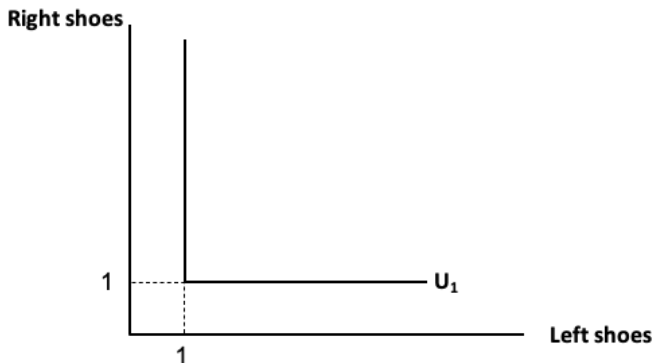
IC $\rightarrow$ Same utility.

Two goods that need to be used together so as to derive utility.



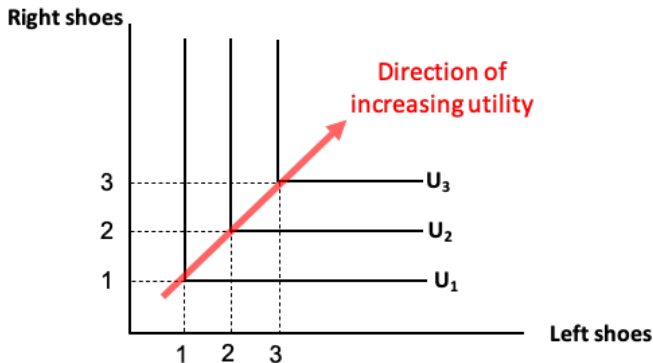
Perfect Complements

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Perfect Complements

Two goods that need to be used together so as to derive utility.

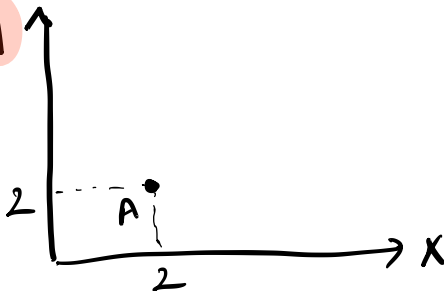


Useless good

Also called neutral good

Try yourself!

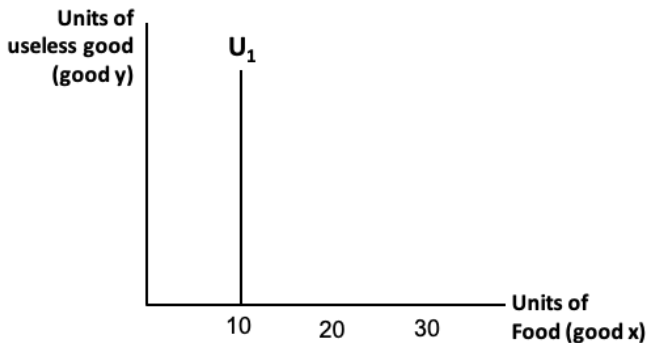
Useless good is a commodity that does not give any utility to the consumer. Let's assume good y is the useless good.



Useless good

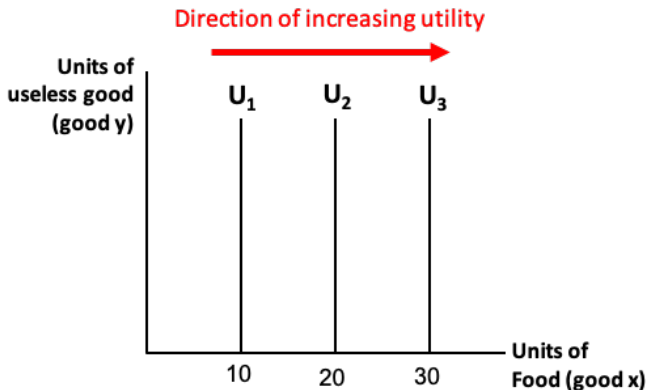
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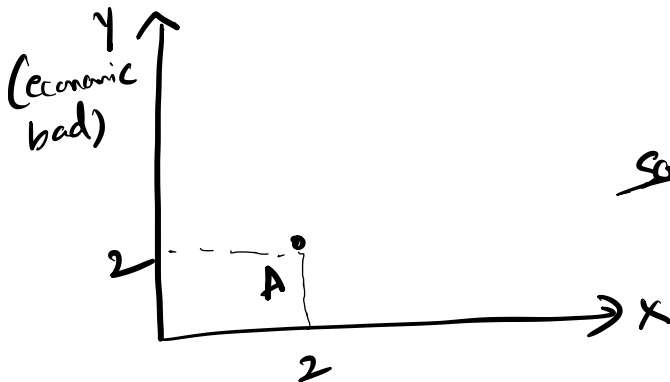


Economic Bad

Try to find the shape of the indifference curve.

Try yourself!
Disutility.

An economic bad is a good that confers negative utility to the consumer. Let's assume good Y is an economic bad.

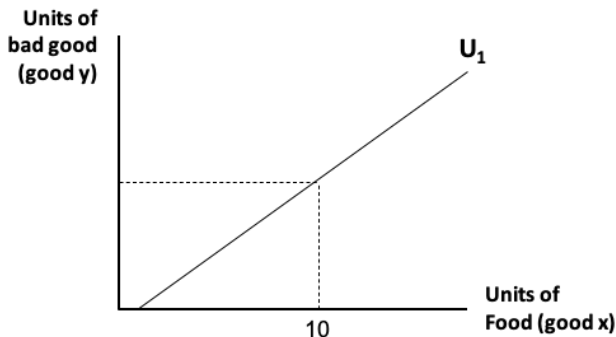


same utility as
A.

Economic Bad

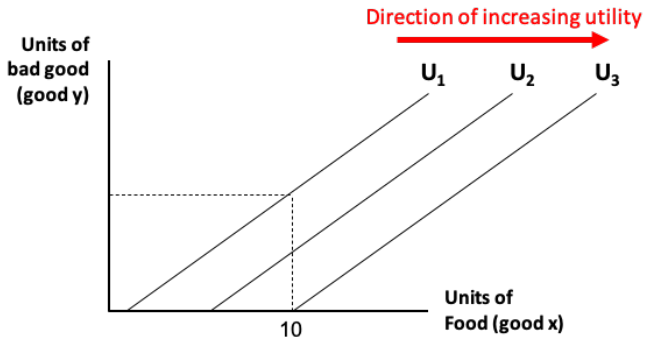
Try to find the shape of the indifference curve.

An economic bad is a good that confers negative utility to the consumer. Lets assume good Y is an economic bad.



Economic Bad

An economic bad is a good that confers negative utility to the consumer.



Preference / Utility.

Budget Constraint

Afford?

The other part of consumer behavior - Budgets!

Budget Constraint (BC)

If, I = Income, $\rightarrow$ numbers
 P_x = Price of good x $\rightarrow$ numbers.
 P_y = Price of good y $\rightarrow$ numbers.

$$\underline{I} = \underbrace{P_x \cdot X}_{\text{Spent on good X}} + \underbrace{P_y \cdot Y}_{\text{Spent on good Y}}$$

Budget constraint

Linear function

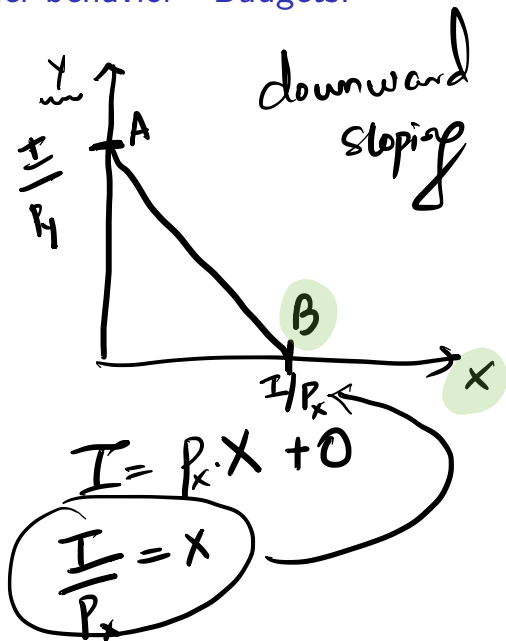
The other part of consumer behavior - Budgets!

Budget Constraint (BC)

If, I = Income,
 P_x = Price of good x
 P_y = Price of good y

✓ The equation of this line is
 $I = P_x \cdot x + P_y \cdot y$

slope = $\frac{-P_x}{P_y}$
-ve



The other part of consumer behavior - Budgets!

Budget Constraint (BC)

If, I = Income,

P_x = Price of good x

P_y = Price of good y

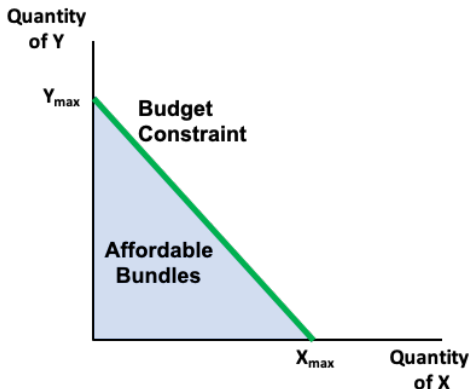
The equation of this line is

$$I = P_x * x + P_y * y$$

The maximum amount of each good that the consumer can purchase is given by,

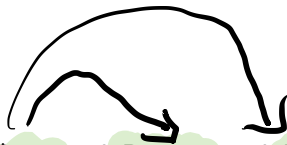
$$Y_{max} = \frac{I}{P_y} \text{ and } X_{max} = \frac{I}{P_x}.$$

$$\text{Slope of BC} = -\frac{P_x}{P_y}$$



Budget constraint

Example



Suppose, Sheela has an income of \$100 and $P_x = 5$, and $P_y = 2$.

(a) What is the maximum amount of goods x and y that she can consume?

$$\frac{100}{5} = 20 \quad \frac{100}{2} = 50$$

(b) What is the equation of her budget constraint?

$$100 = 5 \cdot x + 2 \cdot y$$

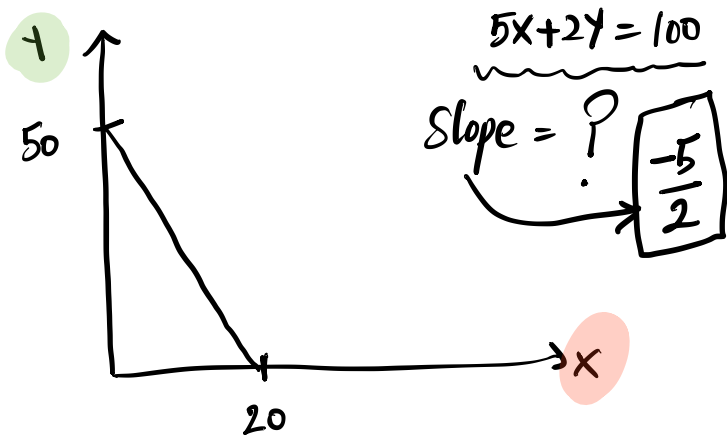
(c) Draw her budget constraint on a graph.

$$5x + 2y = 100$$

Budget constraint

Suppose, Sheela has an income of \$100 and $P_x = 5$, and $P_y = 2$.

- (a) What is the maximum amount of goods x and y that she can consume?
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Budget constraint

Suppose, Sheela has an income of \$100 and $P_x = 5$, and $P_y = 2$.

- (a) What is the maximum amount of goods x and y that she can consume?
- (b) What is the equation of her budget constraint?

Part a

The maximum amount of good X = $X_{max} = \frac{I}{P_x} = \frac{100}{5} = 20$

The maximum amount of good Y = $Y_{max} = \frac{I}{P_y} = \frac{100}{2} = 50$

Part b

The equation of the BC is $100 = 5x + 2y$ ($I = P_x * x + P_y * y$)

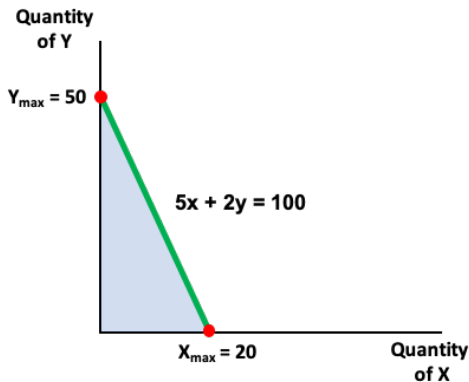
Plotting budget constraint

Solution of Part c – Draw her budget constraint on a graph.

Plotting budget constraint

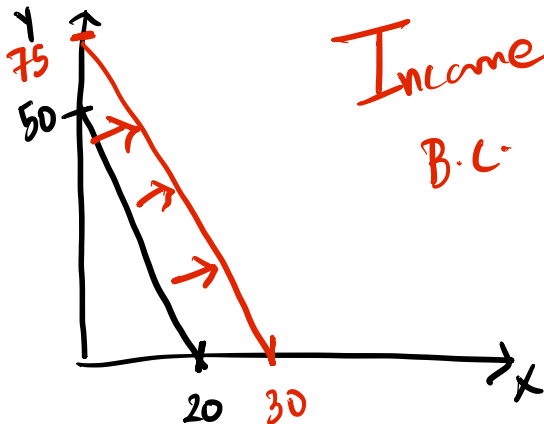
Solution of Part c – Draw her budget constraint on a graph.

In the given example, $X_{max} = 20$, and $Y_{max} = 50$. Connect both the points to get the budget constraint.



Changes in budget constraint

How does BC change if income increases from 100 to 150, while prices remain $P_x = 5$, and $P_y = 2$



Changes in budget constraint

How does BC change if income increases from 100 to 150, while prices remain $P_x = 5$, and $P_y = 2$

Think of what happens to the following:

► **Intercept**

$$Y_{max} = \frac{I}{P_y} \text{ and}$$

$$X_{max} = \frac{I}{P_x}.$$

► **Slope** = $-\frac{P_x}{P_y}$

Changes in budget constraint

How does BC change if income increases from 100 to 150, while prices remain $P_x = 5$, and $P_y = 2$

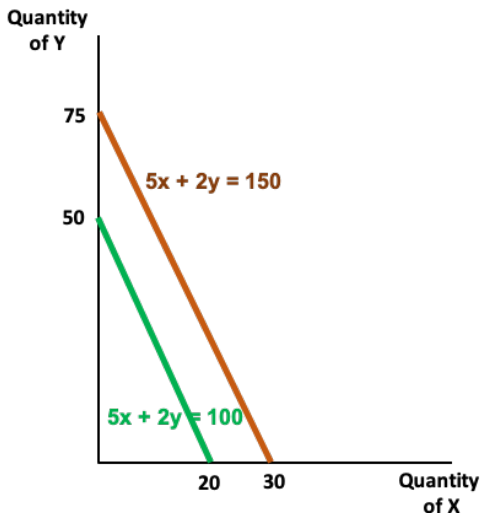
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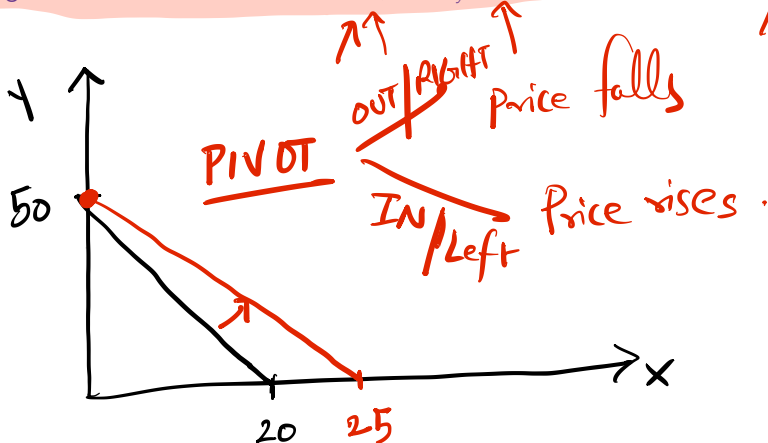
► **Slope** = $-\frac{P_x}{P_y}$



Changes in budget constraint

How does BC change when the price of a good changes. Example: $P_x \downarrow$ from \$5 to \$4.

Other things remain the same – Income = 100, and $P_y = 2$



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Changes in budget constraint

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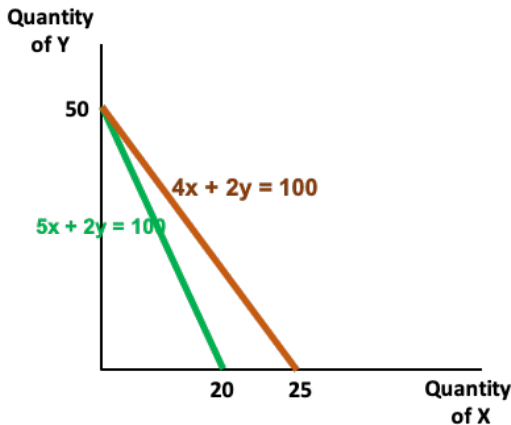
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Determining optimal consumption

Main intuition

Utility / Preferences ✓

Budget constraint ✓

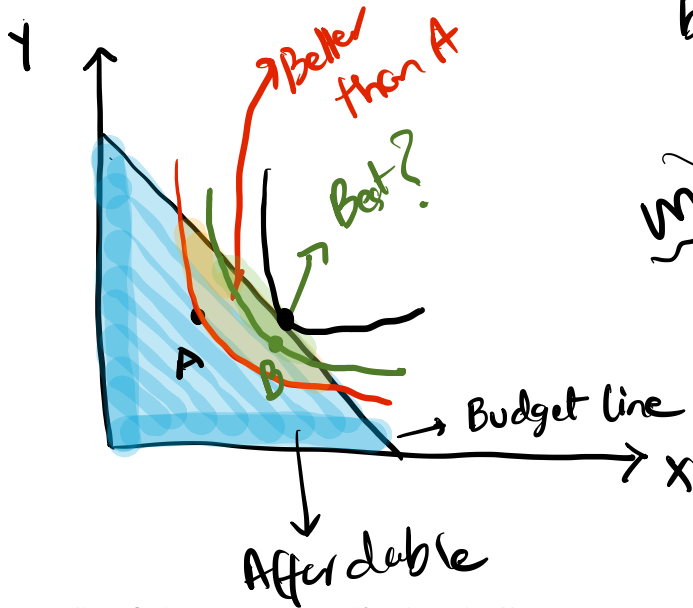
Determining optimal consumption

Main intuition

Individuals are trying to consume a bundle that gives them maximum satisfaction (utility) among the ones that they can afford (within the budget constraint).

Optimal Consumption = A. ?

How do we use IC and budget line to get optimal consumption bundle?

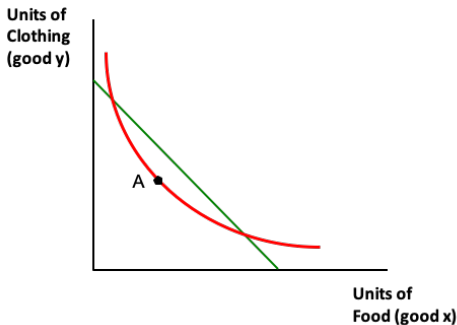


~~Better
bundle~~

~~insigent~~

Optimal Consumption

How do we use IC and budget line to get optimal consumption bundle?

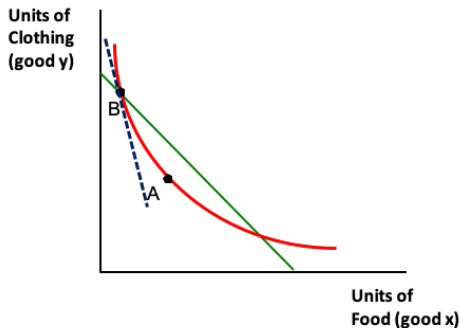


Is point A optimal?

Point A is not optimal as the whole of the budget (income) is not being spent!

Optimal Consumption

Is point B optimal?



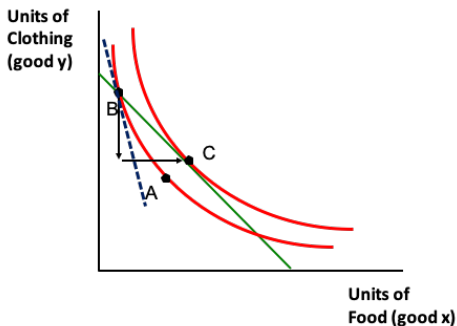
B has the same utility as A, but the whole income is used.

But is point B optimal?

Point B is not optimal as we could potentially reach a higher IC with the same budget constraint.

Optimal Consumption

Is point C optimal?



Point C is optimal as it represents the highest IC that can be reached within the budget.

MRS (slope of IC) = the slope of BC.

Optimal Consumption Bundle

- (i) Use the entire budget, and (ii) slope of budget line = slope of IC

→ MRS

$$-\frac{P_x}{P_y} = -\frac{MU_x}{MU_y}$$

benefit

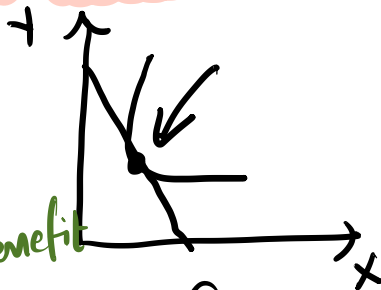
$$\frac{MU_x}{P_x}$$

cost

$$= \frac{MU_y}{P_y}$$

benefit

cost



Barg-for-the
buck
interpretation

Optimal Consumption Bundle

(i) Use the entire budget, and (ii) slope of budget line = slope of IC

$$\underbrace{\text{Slope of the budget line}}_{\frac{-P_x}{P_y}} = \underbrace{\text{Slope of IC}}_{\text{MRS} = \frac{-MU_x}{MU_y}}$$

$$\cancel{\frac{P_x}{P_y}} = \cancel{\frac{MU_x}{MU_y}}$$

$$\frac{P_x}{P_y} = \frac{MU_x}{MU_y}$$

Optimal Consumption Bundle

"Bang for the buck" interpretation

The optimal consumption bundle is generally determined by the point at which

$$\frac{P_x}{P_y} = \frac{MU_x}{MU_y}$$
$$\frac{MU_y}{P_y} = \frac{MU_x}{P_x}$$

Optimal Consumption Bundle

"Bang for the buck" interpretation

The optimal consumption bundle is generally determined by the point at which

$$\begin{aligned}\frac{P_x}{P_y} &= \frac{MU_x}{MU_y} \\ \frac{MU_y}{P_y} &= \frac{MU_x}{P_x} \\ \underbrace{\frac{MU_y}{P_y}}_{\text{Good Y - bang for the buck}} &= \underbrace{\frac{MU_x}{P_x}}_{\text{Good X - bang for the buck}}\end{aligned}$$

Intuition: MU is the benefit of consuming one more unit of good, while P is the cost of that unit. Therefore the "cost-benefit" ratio of the two goods should be the same at the equilibrium point.

Optimal Consumption Bundle

Example

Optimal Consumption Bundle

Example

$$U(x,y) = \sqrt{xy} = x^{1/2} \cdot y^{1/2}$$

Amanda has an income of \$200, and she spends all this income on food (x) and clothing (y). The price of food is \$4 per unit, while the price of clothing is \$5 per unit. Her utility function is given by $u(x,y) = \sqrt{xy}$. Compute the following:

- ▶ Equation and slope of the budget line. $200 = 4x + 5y$
slope = $-\frac{4}{5}$
- ▶ Marginal utilities of the two goods.
- ▶ Slope of the indifference curve. MRS
- ▶ Optimal consumption bundle.
- ▶ Optimal consumption bundle if price of clothing changes to \$2.

$P_x: 4 \rightarrow 2$ Try yourself!

Solution

Amanda has an income of \$200, and she spends all this income on food (x) and clothing (y). The price of food is \$4 per unit, while the price of clothing is \$5 per unit. Her utility function is given by $u(x, y) = \sqrt{xy}$.

$$MU_x = \frac{\partial U}{\partial x} = \frac{\partial}{\partial x} [x^{1/2} \cdot y^{1/2}] = y^{1/2} \cdot \frac{\partial}{\partial x} [x^{1/2}] = y^{1/2} \cdot \frac{1}{2} x^{-1/2}$$

$$MU_y = \frac{\partial U}{\partial y} = \frac{\partial}{\partial y} [x^{1/2} \cdot y^{1/2}] = x^{1/2} \cdot \frac{\partial}{\partial y} [y^{1/2}] = x^{1/2} \cdot \frac{1}{2} y^{-1/2}$$

$$MRS = \frac{-MU_x}{MU_y} = \frac{-y^{1/2} \cdot \frac{1}{2} x^{-1/2}}{x^{1/2} \cdot \frac{1}{2} y^{-1/2}} = \frac{-y^{1/2} \cdot y^{1/2}}{x^{1/2} x^{1/2}} = \frac{-y}{x}$$

① Budget exhausted: $200 = 4x + 5y$

② slope of B.C. = slope of I.C.

$$\frac{-4}{5} = \frac{-y}{x} \Rightarrow y = \frac{4x}{5}$$

$$y = \frac{4(25)}{5} = 20$$

$$\begin{aligned} 200 &= 4x + 5y \\ 200 &= 4x + 5\left(\frac{4x}{5}\right) \end{aligned}$$

$$\begin{aligned} 200 &= 4x + 4x \\ x &= 25 \end{aligned}$$

Solution

Amanda has an income of \$200, and she spends all this income on food (x) and clothing (y). The price of food is \$4 per unit, while the price of clothing is \$5 per unit. Her utility function is given by $u(x, y) = \sqrt{xy}$.

new $P_x = 2$.

① Budget is exhausted: $200 = 2x + 5y$

② slope of B.C. = slope of I.C

$$-\frac{2}{5} = -\frac{y}{x}$$

$$y = \frac{2x}{5}$$

$$y = \frac{2(50)}{5} = 20$$

$$200 = 2x + 5y$$

$$200 = 2x + 5\left(\frac{2x}{5}\right)$$

$$200 = 2x + 2x$$

$$x = 50$$